ED-EQ coupling in nanoparticle arrays

This set of notes concerns the papers:
Metasurfaces with Electric Quadrupole and Magnetic Dipole Resonant Coupling (https://pubs.acs.org/doi/pdf/10.1021/acsphotonics.7b01520?casa_token=v04BC17EN1UAAAAA:rnwHOYYYx3r5GktBJd6xuvCDCUcEA9GfVKDMCcOtKI6fIYkbz4y1dhSyHuZg-VkLFQ-OJosvvp2ULT4) by **Babicheva and Evlyukhin** (**Reference 1**), Optical response features of Si-nanoparticle arrays (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.82.045404) (**Reference 2**), Colloquium: Light scattering by particle and hole arrays (https://journals.aps.org/rmp/pdf/10.1103/RevModPhys.79.1267) by **de-Abajo** (**Reference 3**)

Summary of these notes

In **Section 1**, we formulate the dipole lattice sum in the vein of **Reference 3**. In **Section 2**, we show that the dipole sum diverges at particular values of the wavelength, and we find the strength of this divergence in the case of a square lattice. In **Sections 3 and 4**, we find the effect of including magnetic multipole and electric dipole contributions. In **Section 5**, we prove an important identity concerning the field of a magnetic dipole (this identity is also used in **Section 3**). Lastly, in **Section 7**, we find a closed form solution for the imaginary part of the lattice Green's function sum, deriving a key result from **Reference 3**.

Section 1: Dipole lattice sum (de-Abajo)

Figure 6 (a) depicts a curve of the quantity $a^3\mathbf{G}_{xx}$. Note that de-Abajo's definition of the Green's function differs from the usual definition (e.g **Equation 2.105** in **Novotny and Hecht**) by a factor of $4\pi k^2$. The factor of 4π comes from a difference in unit systems and the factor of k^2 comes from the fact that the usual definition of the Dyadic Green's function is that $k^2\mathbf{G}(\mathbf{r},\mathbf{r}_0)\mathbf{p}(\mathbf{r}_0) = \mathbf{E}(\mathbf{r})$, while de Abajo subsumes the factor of k^2 into the definition of the Green's tensor, so that that Green's tensor directly connects the dipole moment to its induced field. Therefore, the quantity that he plots is actually given by:

$$a^3 \sum_{\mathbf{r}} \frac{(kr)^2 - 1 + i(kr) + 3x^2/r^2 - 3i(kx)(x/r) - (kx)^2}{r^3} e^{ikr},$$

where $r \equiv |\mathbf{r}|$, and $\mathbf{r} = a(n,m)$, with a being the lattice constant, and $x = an$. If we define $\tan^{-1}(m/n) = \Theta_{nm}$, then we obtain the following:

$$\sum_{n,m} \frac{(kr)^2 \sin^2(\Theta_{nm}) + \Big[3\cos^2(\Theta_{nm}) - 1\Big]\Big[1 - ikr\Big]}{(r/a)^3} e^{ikr},$$

with the understanding that $k = (2\pi/\lambda)$, $r = a\sqrt{n^2 + m^2}$, which gives us $kr = (2\pi a/\lambda)\sqrt{n^2 + m^2}$. Numerically, it seems quite hard to converge this quantity to obtain de-Abajo's result in **Figure 6**. For each lattice sum, one seems to need to sum about $\approx 10^8$ points to achieve convergence.

Section 2: Divergence of the lattice sum for in-plane polarization

In this section, we wish to prove **Equation 13** of **Reference 3**. This result gives the divergence of the lattice sum, $4\pi k^2 a^3 \mathbf{G}_{xx}$ (the quantity $4\pi k^2 \mathbf{G}_{xx}$ is given by the lattice sum of **Equation 1** of **Reference 3**). This result gives the divergence of the lattice Green's function sum as the wavelength of light approaches the lattice constant (**for a square lattice**; i.e. $a \equiv L_x = L_y$). In order to find the nature of this divergence, we use **Equation A7** of **Reference 2** (or one may equivalently use **Equation 7** of **Reference 3**). Explicitly, we have:

$$\begin{aligned} 4\pi k^2 a^3 \mathbf{G}_{xx} = \frac{i(4\pi k^2 a^3)}{2a^2k^2} \sum_{m_x,m_y} \frac{k^2 - (2\pi m_x/L_x)^2}{\sqrt{k^2 - (2\pi)^2[(m_x/L_x)^2 + (m_y/L_y)^2]}} = \\ 2\pi a \sum_{m_x,m_y} \frac{k^2 - (2\pi m_x/a)^2}{\sqrt{(2\pi)^2[(m_x/a)^2 + (m_y/a)^2] - k^2}}, \end{aligned}$$

where $k \equiv 2\pi/\lambda$ is the wavevector, m_x, m_y index the reciprocal lattice vectors. When the wavelength approaches the lattice constant, i.e. $k \approx 2\pi/a$, the divergent terms are those in the lattice sum corresponding to $m_x = 0, m_y = \pm 1$, of which there are two. We note that the terms corresponding to $m_x = \pm 1, m_y = 0$ do not contribute to the divergence due to the numerator of the summand. Near the wavelength, we may make the replacement $(2\pi/a)^2 - k^2 \approx (4\pi/a)(2\pi/a - 2\pi/\lambda)$. This allows us to express the Green's function sum as:

$$4\pi k^2 a^3 \mathbf{G}_{xx} \approx 4\pi a \frac{(2\pi/a)^2}{\sqrt{(2\pi/a)^2 - k^2}} \approx \frac{(4\pi a)(2\pi/a)^2}{\sqrt{(4\pi/a)(2\pi/\lambda)(\lambda/a - 1)}} \approx \frac{2\sqrt{8}\pi^2}{\sqrt{\lambda/a - 1}} = \frac{4\sqrt{2}\pi^2}{\sqrt{\lambda/a - 1}},$$

which is precisely the result cited in **Equation 13** of **Reference 3**.

Section 3: Inclusion of higher order moments in the lattice sum

In this section, we wish to prove the set of four relations in the second unnumbered equation after **Equation 3** in **Reference 2**. For notational simplicity, we restrict to the case where the environment is vacuum or air, i.e. $k \equiv k_0 \equiv k_d$ (where k_0, k_d are defined in **Reference 2**). The electric dipole fields are more-or-less obvious. We know that the electric field due to an oscillating electric dipole is given by $(-i\omega)(i\omega\mu_0)\mathbf{G}(\mathbf{r}, \mathbf{r}_0) \cdot \mathbf{p}(\mathbf{r}_0)$, where $\mathbf{G}(\mathbf{r}, \mathbf{r}_0)$ is the Dyadic Green's function. Therefore,we may immediately write for the field of a dipole located at $\mathbf{r}_0$ and evaluated at $\mathbf{r}$:

$$\mathbf{E}_\mathbf{p}(\mathbf{r}) = \omega^2(1/c^2\varepsilon_0)\mathbf{G}(\mathbf{r}, \mathbf{r}_0) \cdot \mathbf{p}(\mathbf{r}_0) = (k^2/\varepsilon_0)\mathbf{G}(\mathbf{r}, \mathbf{r}_0) \cdot \mathbf{p}(\mathbf{r}_0),$$

which is the first of the four relations. The corresponding magnetic field may be easily calculated from the electric field calculated above:

$$(k_0^2/\varepsilon_0)\nabla \times \mathbf{G}(\mathbf{r}, \mathbf{r}_0) \cdot \mathbf{p}(\mathbf{r}_0) = i\omega\mu_0 \mathbf{H}_\mathbf{p}(\mathbf{r}) \rightarrow \mathbf{H}_\mathbf{p}(\mathbf{r}) = -i\omega\nabla \times \mathbf{G}(\mathbf{r}, \mathbf{r}_0) \cdot \mathbf{p}(\mathbf{r}_0),$$

where the curl is with respect to the observation coordinate, $\mathbf{r}$. We may write this slightly differently, by using $\omega = ck$, as:

$$\mathbf{H}_\mathbf{p}(\mathbf{r}) = (kc/i)\mathbf{G}(\mathbf{r}, \mathbf{r}_0)\nabla \times \Big[\mathbf{G}(\mathbf{r}, \mathbf{r_0}) \cdot \mathbf{p}(\mathbf{r}_0)\Big],$$

which is the second of the four relations. The last two of the four relations concern fields radiated from a magnetic dipole. In order to derive these, we write the vector potential due to a magnetic dipole (in the Lorenz gauge; we read this off from **Equation 9.33** of **Jackson Electrodynamics**):

$$\begin{aligned}\mathbf{A}_{\mathbf{m}}(\mathbf{r},\omega) = \frac{ik\mu_0}{4\pi}(\mathbf{r}\times\mathbf{m})\frac{e^{ikr}}{r^2}\left[1-\frac{1}{ikr}\right] \rightarrow \\ \mathbf{H}_{\mathbf{m}}(\mathbf{r},\omega) = \frac{ik}{4\pi}\nabla\times\left[(\mathbf{r}\times\mathbf{m})\frac{e^{ikr}}{r^2}\left[1-\frac{1}{ikr}\right]\right],\end{aligned}$$

where, for simplicity, we set the location of the magnetic dipole, $\mathbf{r}_0$ to the origin. We note that this expression is of the form

$$\alpha\Big[\nabla\times(\mathbf{r}\times\mathbf{m})f(r)\Big]_i = \alpha\varepsilon_{kij}\varepsilon_{klm}\partial_j(\mathbf{r}_l\mathbf{m}_m f(r)),\ \text{with}$$
$$f(r)\equiv\frac{e^{ikr}}{r^2}\left[1-\frac{1}{ikr}\right],\ \text{and}\ \alpha\equiv\frac{ik}{4\pi}$$

To simplify this expression, we use identities of the Levi-Cevita tensor, giving us:

$$\begin{aligned}\varepsilon_{kij}\varepsilon_{klm}\partial_j(\mathbf{r}_l\mathbf{m}_m f(r)) = (\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl})\partial_j(\mathbf{r}_l\mathbf{m}_m f(r)) = \\ \partial_j(\mathbf{r}_i\mathbf{m}_j f(r)) - \partial_j(\mathbf{r}_j\mathbf{m}_i f(r)) = \\ \mathbf{m}_i f(r) + f'(r)(\mathbf{r}\cdot\mathbf{m})(\mathbf{r}_i/r) - 3\mathbf{m}_i f(r) - \mathbf{r}_j\mathbf{r}_j\mathbf{m}_i f'(r)/r = \\ -(2f(r)+rf'(r))\mathbf{m} + (f'(r)/r)(\mathbf{r}\cdot\mathbf{m})\mathbf{r} = \\ \Big[-(2f(r)+rf'(r))\delta_{ij} + (f'(r)/r)\mathbf{r}_i\mathbf{r}_j\Big]\mathbf{m}_j\end{aligned}$$

At this point, we need only calculate the quantity, $f'(r)$:

$$\partial_r\left[e^{ikr}\left[\frac{1}{r^2}-\frac{1}{ikr^3}\right]\right] = e^{ikr}\left[\frac{ik}{r^2}-\frac{1}{r^3}-\frac{2}{r^3}+\frac{3}{ikr^4}\right] = e^{ikr}\left[\frac{ik}{r^2}-\frac{3}{r^3}-\frac{3i}{kr^4}\right]$$

From this, we readily obtain:

$$-(2f(r) + rf'(r))\delta_{ij} + (f'(r)/r)\mathbf{r}_i\mathbf{r}_j = -e^{ikr}\left[\left[\frac{ik}{r} - \frac{1}{r^2} - \frac{i}{kr^3}\right]\delta_{ij} + \left[\frac{ik}{r} - \frac{3}{r^2} - \frac{3i}{kr^3}\right]\mathbf{e_r}\mathbf{e_r}\right],$$

where we introduced the unit vectors, $\mathbf{e_r} \equiv \mathbf{r}/r$. Note the similarity between the above and the more familiar Dyadic Green's function. In fact, we immediately see that the magnetic field may be written as:

$$\mathbf{H}(\mathbf{r}) = \frac{ik}{4\pi}e^{ikr}\left[\frac{i}{kr^3}\left(1 - i(kr) - (kr)^2\right)\delta_{ij} + \frac{i}{kr^3}\left(-3 + 3i(kr) + (kr)^2\right)\mathbf{e_r}\mathbf{e_r}\right]\mathbf{m} = k^2\mathbf{G}(\mathbf{r},\mathbf{r}_0)\mathbf{m}(\mathbf{r}_0),$$

which is the third of the four relations. Finally, we find the electric field corresponding to the magnetic dipole:

$$\mathbf{E}(\mathbf{r},\omega) = \frac{ik^2}{\omega\varepsilon_0}\nabla\times\mathbf{G}(\mathbf{r},\mathbf{r}_0)\mathbf{m}(\mathbf{r}_0) = \frac{ik}{c\varepsilon_0}\nabla\times\mathbf{G}(\mathbf{r},\mathbf{r}_0)\mathbf{m}(\mathbf{r}_0) = \frac{ik}{c\varepsilon_0}\mathbf{g}(\mathbf{r},\mathbf{r}_0)\times\mathbf{m}(\mathbf{r}_0),$$

which is the last of the four relations. Note that the quantity $\mathbf{g}(\mathbf{r},\mathbf{r}_0)$ is introduced in **Reference 1** and we prove the last equality in **Section 5** of this document.

Section 4: Inclusion of quadrupole in lattice sum

In this section, we wish to formulate the effect of including higher order multipole moments in the lattice site interactions. The primary point of reference for this section is **Reference 1**. In particular, in this section we wish to calculate the exact field distribution of an electric quadrupole (valid in all regimes: near-field and far-field) and ensure that the obtained result is consistent with **Equation 1** of **Reference 1**. Our starting point is **Equation 9.38** of **Jackson Electrodynamics**. Using the function, $f(r)$, which we defined in the previous section, we may write the magnetic field $\mathbf{H_Q}(\mathbf{r},\omega)$ (or $\mathbf{B_Q}(\mathbf{r},\omega)$) as follows (note that we use the notation of **Jackson**):

$$\mathbf{H_Q}(\mathbf{r},\omega) = -\frac{\mu_0 c k^2}{8\pi}\nabla\times f(r)\int(\mathbf{r}\cdot\mathbf{x}')\mathbf{x}'\rho(\mathbf{x}')\mathrm{d}^3\mathbf{x}'$$
$$= -\frac{\mu_0 c k^2}{8\pi}\varepsilon_{ijk}\int\partial_j\Big[f(r)(\mathbf{r}\cdot\mathbf{x}')\mathbf{x}'_k\Big]\rho(\mathbf{x}')\mathrm{d}^3\mathbf{x}'$$
$$= -\frac{\mu_0 c k^2}{8\pi}\varepsilon_{ijk}\int\Big[f(r)\mathbf{x}'_j\mathbf{x}'_k + f'(r)(\mathbf{r}\cdot\mathbf{x}')\mathbf{x}'_k\mathbf{r}_j/r\Big]\rho(\mathbf{x}')\mathrm{d}^3\mathbf{x}'$$

The first term does not contribute because $\mathbf{x}'_j\mathbf{x}'_k$ is symmetric while ε_{ijk} is antisymmetric (equivalently, it corresponds to a cross product of a vector with itself). Furthermore, in order to simplify the second term, we note that

$$\varepsilon_{ijk}f'(r)(\mathbf{r}\cdot\mathbf{x}')\mathbf{x}'_k\mathbf{r}_j/r = \Big[\mathbf{r}\times f'(r)\mathbf{x}'\mathbf{x}'\cdot(\mathbf{r}/r)\Big]_i$$

Using our previously computed form for $f'(r)$:

$$f'(r) = e^{ikr}\left[\frac{ik}{r^2} - \frac{3}{r^3} - \frac{3i}{kr^4}\right] = \frac{(ik)e^{ikr}}{r^2}\left[1 + \frac{3i}{(kr)} - \frac{3}{(kr)^2}\right],$$

we may write down the magnetic field immediately as:

$$\mathbf{B_Q}(\mathbf{r},\omega) = \mu_0\mathbf{H_Q}(\mathbf{r},\omega) = -\frac{\mu_0 c k^2(ik)}{24\pi r^2}e^{ikr}\left[1 + \frac{3i}{(kr)} - \frac{3}{(kr)^2}\right]\mathbf{r}\times(\mathbf{Q}\cdot\mathbf{r}/r),$$

where $\mathbf{Q}$ is the quadrupole tensor. Note that the quadrupole tensor (as defined in, say, **Equation 9.41** of **Jackson**) has, in addition to a term proportional to $\mathbf{x}'\mathbf{x}'$, a term proportional to the unit Dyadic. However, this is fine since that term does not contribute (since it will give a term proportional to the cross product of $\mathbf{r}$ with itself. In order to verify that our answer is consistent with **Reference 1**, we write our result in their notation (see **Equation 5** of **Reference 1**):

$$\mathbf{B_Q}(\mathbf{r},\omega) = (-i\mu_0 ck)\mathbf{q}\times(\mathbf{Q}\cdot\mathbf{n}) = \mu_0 k^2\left[\frac{c}{ik}\right]\mathbf{q}\times(\mathbf{Q}\cdot\mathbf{n})$$

Note that this is entirely consistent with **Equation 1** from **Reference 1**. From this, we may also obtain the electric field:

$$\mathbf{E}_{\mathbf{Q}}(\mathbf{r},\omega) = \nabla \times \left[\frac{k^2}{24\pi r^3\varepsilon_0}e^{ikr}\left[1+\frac{3i}{(kr)}-\frac{3}{(kr)^2}\right]\mathbf{r}\times(\mathbf{Q}\cdot\mathbf{r})\right]$$

At this point, we define a function, $g(r)$, and a constant, β, given by the following two expressions:

$$g(r) \equiv \frac{e^{ikr}}{r^3}\left[1+\frac{3i}{(kr)}-\frac{3}{(kr)^2}\right],\ \beta \equiv \frac{k^2}{24\pi\varepsilon_0}$$

With these two definitions, we may write the field as follows:

$$\begin{aligned}\mathbf{E}(\mathbf{r},\omega) = \beta\nabla\times\left[g(r)(\mathbf{r}\times\mathbf{Q}\cdot\mathbf{r})\right] &= \beta\varepsilon_{kij}\varepsilon_{klm}\left[g(r)\partial_j(\mathbf{r}_l\mathbf{Q}_{m\alpha}\mathbf{r}_\alpha)+g'(r)(\mathbf{r}_l\mathbf{Q}_{m\alpha}\mathbf{r}_\alpha)\frac{\mathbf{r}_j}{r}\right]\\ &= \beta(\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl})\left[(\delta_{jl}\mathbf{Q}_{m\alpha}\mathbf{r}_\alpha+\delta_{j\alpha}\mathbf{r}_l\mathbf{Q}_{m\alpha})g(r)+g'(r)(\mathbf{r}_l\mathbf{Q}_{m\alpha}\mathbf{r}_\alpha)\frac{\mathbf{r}_j}{r}\right]\\ &= \beta\left[\left[\mathbf{Q}_{\alpha\alpha}\mathbf{r}_i-3\mathbf{Q}_{i\alpha}\mathbf{r}_\alpha\right]g(r)+g'(r)\left[\mathbf{r}_i\mathbf{r}_m\mathbf{Q}_{m\alpha}\mathbf{r}_\alpha/r-r\mathbf{Q}_{i\alpha}\mathbf{r}_\alpha\right]\right],\end{aligned}$$

where, in the above, we used the fact that $\delta_{jl}\delta_{jl}=3$. At this point, we also note that the trace of the quadrupole tensor, $\mathbf{Q}_{\alpha\alpha}$ vanishes, which means that, definining a unit vector, $\mathbf{n}=\mathbf{r}/r$, gives us the following expression for the electric field:

$$\mathbf{E}(\mathbf{r},\omega) = \beta\left[r^2g'(r)\left[\mathbf{n}\mathbf{n}\cdot\mathbf{Q}\cdot\mathbf{n}-\mathbf{1}\cdot\mathbf{Q}\cdot\mathbf{n}\right]-3rg(r)(\mathbf{1}\cdot\mathbf{Q}\cdot\mathbf{n})\right]$$

At this point, we calculate $r^2g'(r)$:

$$\begin{aligned}r^2\partial_r\left[e^{ikr}\left[\frac{1}{r^3}+\frac{3i}{kr^4}-\frac{3}{k^2r^5}\right]\right] &= e^{ikr}\left[\frac{ik}{r}-\frac{3}{r^2}-\frac{3i}{kr^3}-\frac{3}{r^2}-\frac{12i}{kr^3}+\frac{15}{k^2r^4}\right]\\ &= e^{ikr}\left[\frac{ik}{r}-\frac{6}{r^2}-\frac{15i}{kr^3}+\frac{15}{k^2r^4}\right] = \frac{ik}{r}e^{ikr}\left[1+\frac{6i}{(kr)}-\frac{15}{(kr)^2}-\frac{15i}{(kr)^3}\right]\end{aligned}$$

We furthermore have:

$$-(r^2g'(r)+3rg(r)) = -\frac{ik}{r}e^{ikr}\left[1+\frac{6i}{kr}-\frac{15}{(kr)^2}-\frac{15i}{(kr)^3}-\frac{3i}{kr}+\frac{9}{(kr)^2}+\frac{9i}{(kr)^3}\right] = -\frac{ik}{r}e^{ikr}\left[1+\frac{3i}{kr}-\frac{6}{(kr)^2}-\frac{6i}{(kr)^3}\right]$$

Therefore, the electric field is given by:

$$\mathbf{E}(\mathbf{r},\omega) = \frac{ik^3e^{ikr}}{24\pi\varepsilon_0 r}\left[\left[1+\frac{6i}{kr}-\frac{15}{(kr)^2}-\frac{15i}{(kr)^3}\right](\mathbf{nn}\cdot\mathbf{Q})+\left[-1-\frac{3i}{kr}+\frac{6}{(kr)^2}+\frac{6i}{(kr)^3}\right](\mathbf{1}\cdot\mathbf{Q})\right]\cdot\mathbf{n}$$

Note that this is also consistent with the result of **Equation 1** of **Reference 1**.

Section 5: Proof of relation after Equation 3 of Reference 2

Reference 2 has a very neat result, which we wish to prove (we use the notation of **Reference 2** and note that the relation which we wish to prove may be found just after **Equation 3** of the paper):

$$\nabla\times\mathbf{G}(\mathbf{r}_l,\mathbf{r}_j)\mathbf{p}_j = \mathbf{g}(\mathbf{r}_l,\mathbf{r}_j)\times\mathbf{p}_j.$$

We first note that

$$\mathbf{G} = (\delta_{ij}+k^{-2}\nabla_i\nabla_j)(e^{ikr}/4\pi r) \to \varepsilon_{kli}\mathbf{p}_j\nabla_l(\delta_{ij}+k^{-2}\nabla_i\nabla_j)(e^{ikr}/4\pi r) = \nabla(e^{ikr}/4\pi r)\times\mathbf{p} = \frac{e^{ikr}}{4\pi r^2}\left[ik-\frac{1}{r}\right](\mathbf{r}\times\mathbf{p})\equiv\mathbf{g}\times\mathbf{p},$$

as desired. Note that we discarded a term proportional to $\varepsilon_{kli}\nabla_l\nabla_i$, since this corresponds to the curl of a gradient (which vanishes).

Section 6: Reflection/transmission in the absence of coupling

We consider a dipole array in the absence of coupling. Note that in this section, we sometimes use $\Omega = a^2$ to denote the unit cell volume and R for the radius of the spherical scatterers under consideration. Using **Equations 20 and 21** of **Reference 2**, we have that the reflection/transmission coefficients will be given by (note that here we define our polarizabilities by $\alpha \to \alpha/\varepsilon_0$ so that the polarizability has units of volume):

$$r = \frac{ik\alpha}{2\Omega}, t = 1 + \frac{ik\alpha}{2\Omega} \to |r|^2 + |t|^2 = \frac{k^2|\alpha|^2}{2\Omega^2} + \frac{ik(\alpha - \alpha^*)}{2\Omega} + 1 \to \\ \frac{k|\alpha|^2}{\Omega} + i(\alpha - \alpha^*) < 0 \to (k|\alpha|^2/2\Im\alpha) < \Omega,$$

which sets a minimum lattice spacing for the validity of the dipole approximation (in the absence of coupling). Note that $\Im\alpha/|\alpha|^2 = -\Im(1/\alpha)$, so we may write the above condition as:

$$-\Im(1/\alpha)\Omega > k/2 \to (ka)^2/3\pi > 1 \to a^2 > 3\pi\lambda^2/(2\pi)^2 \to a > \lambda/2,$$

where we used $\Im(1/\alpha) = -k^3/6\pi$, for a sphere with a lossless dielectric function, and the last inequality is only approximate. Therefore, in the uncoupled regime, we predict that the dipole method is valid for all wavelengths $\lambda < 2a$. This is actually seemingly quite close to the true result in the presence of coupling (see **Figure 8** of **Reference 2**).

Let us now consider the CPA condition: $r + t = 0$. This corresponds to the condition:

$$\frac{k\alpha}{\Omega} = i \to -ik/\Omega = 1/\alpha = \frac{1}{R^3}\left[\frac{1}{4\pi}\frac{\varepsilon_m + 2}{\varepsilon_m - 1} - i(kR)^3/6\pi\right]$$

In order to satisfy this condition, we require $\Re(\varepsilon) = -2$, and we obtain:

$$\frac{k}{\Omega} = \frac{1}{R^3}\left[\frac{\varepsilon''}{12\pi} + \frac{(kR)^3}{6\pi}\right] \to 12\pi(kR)(R^2/\Omega) - 2(kR)^3 = \varepsilon''$$

Note that we have a threshold set by (from requiring that the imaginary part of the dielectric constant be strictly positive):

$$6\pi > (ka)^2 \to 3\pi/(2\pi^2) > (a/\lambda)^2 \to (\lambda/a)^2 > 2\pi/3$$

Section 7: Closed form solution to imaginary part of Green's function sum.

de-Abajo's paper (**Reference 3**) has a quite nice formulation (**Equation 10**) for the imaginary part of the lattice sum, which we seek to derive here. We first note that the Green's function for one scatterer (in de-Abajo's notation and units) is given by:

$$\mathrm{G}_{xx} = (k^2 + \partial_x^2)(e^{ikr}/r) \rightarrow \Im G_{xx} = (k^2 + \partial_x^2)(\sin(kr)/r),$$

For a general function of r, which we denote by $f(r)$, we may write

$$\partial_x f(r) = (x/r) f'(r) \rightarrow \partial_x^2 f(r) = (1/r) f'(r) + (x/r)^2 f''(r) - (x^2/r^3) f'(r) = \frac{f'(r)}{r}(1 - (x/r)^2) + (x/r)^2 f''(r)$$

In our case, for which we have $f(r) \equiv \sin(kr)/r$, we have

$$\partial_r f(r) = \frac{(kr)\cos(kr) - \sin(kr)}{r^2} \approx \frac{(kr) - (kr)^3/2 - (kr) + (kr)^3/6}{r^2} = -k^3 r/3, \quad \partial_r^2 f(r) \approx -k^3/3$$

Note that since $\partial_r f(r) \propto r$, we have that $\partial_r^2 f(r) = f(r)/r$ (in the limit $r \rightarrow 0$). Therefore, we have that the imaginary part of the Green's function is given by:

$$\Im G_{xx} \approx k^3 - k^3/3 = (2/3)k^3$$

Therefore, in the lattice sum, we subtract this from the sum over all lattice points. The first term in **Equation 10** comes from the observation that (in de-Abajo's notation) $k_z^g = \sqrt{(2\pi/\lambda)^2 - (2\pi m/a)^2}$ and if $\lambda > a$, then for any $m \neq 0$, we have k_z^g purely imaginary. This means that the only term that will contribute to the imaginary part of the lattice Green's function (see **Equation 7 of Reference 3**) will correspond to $m = 0$ (the zeroth order diffraction term).